Exploring Algebraic Inequalities

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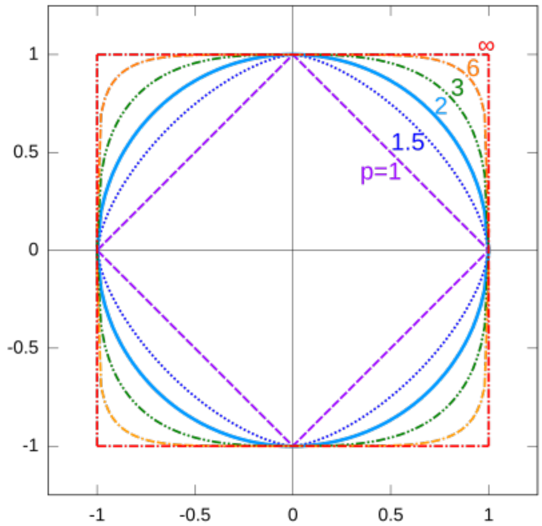

1 Introduction

This article follows what I taught in Senior Math Team regarding algebraic inequalities. I was initially interested in how, often, many inequalities can be proven using a variety of methods, so I was curious of the interdependence of the fundamental inequalities we encounter in competition math. It appears that a lot of what we've seen relies on Jensen's, with the concept of convexity, and this extends in many ways. In my lesson, I also explored extending the Cauchy-Schwarz inequality to Hölder's inequality, and its implication when we consider L^p spaces.

2 Convexity and Jensen's Inequality

For a twice-differentiable real valued function $f(x)$, we define it to be **convex** if $f''(x) \geq 0$. This definition is probably familiar to a calculus student, but this is only one definition. There are multiple equivalent definitions of convexity for functions, and the concept of convexity extends to sets of points as well. Let's convert this definition of convexity to a more broadly used one. Specifically, we'll show that $f''(x) \geq 0$ on an interval I is equivalent to the statement that the secant line between $(x_1, f(x_1))$ and $(x_2, f(x_2))$ with $x_1, x_2 \in I$ (and $x_1 < x_2$) lies above the function $f(x)$.

Assume on some interval I we have $f''(x) > 0$ for all $x \in I$. The slope of the secant line from x_1 to x_2 is

$$m = \frac{f(x_2) - f(x_1)}{x_2 - x_1}.$$

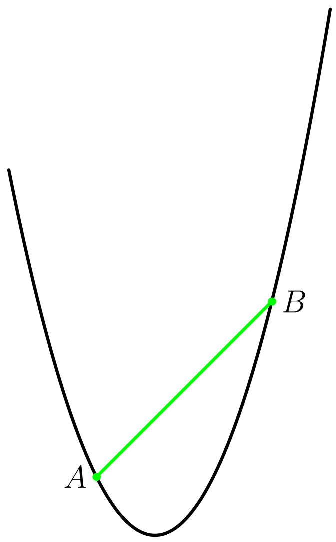

Figure 1: Convex quadratic equation ($f(x) = ax^2 + bx + c$ with $a > 0$)

Then, the equation of the secant line is

$$h(x) = f(x_1) + m(x - x_1).$$

We want to prove that $h(x) > f(x)$ on the interval $[x_1, x_2]$. Let

$$g(x) = f(x) - h(x) = f(x) - (f(x_1) + m(x - x_1)).$$

Then, $g'(x) = f'(x) - m$ and $g''(x) = f''(x) \geq 0$. By the mean value theorem, there exists $a \in [x_1, x_2]$ such that $f'(a) = m$, so $g(x)$ reaches a local extremum at a since $g'(a) = 0$. Since $g''(x) \geq 0$, we know $g'(x)$ is nondecreasing so g reaches a local minimum. Finally, since $g(x_1) = g(x_2) = 0$, this means that $g(x) \leq 0$ for $x \in [x_1, x_2]$, so the secant line always lies above the convex function $f(x)$. Note that it is impossible to have $g(x) > 0$ as this would imply $g(x)$ has a local maximum, but since $g'(x)$ is nondecreasing this is impossible.

Let's try to state this result more formally. We can parameterize the points on the secant line between x_1 and x_2 with parameter $0 \leq t \leq 1$, so any x-coordinate on the line segment is of the form

$$x = x_1 t + x_2(1 - t).$$

Since $f(x) \leq h(x)$, we have

$$\begin{aligned} f(x_1 t + x_2(1-t)) &\leq h(x_1 t + x_2(1-t)) \\ &= f(x_2) + \frac{f(x_2) - f(x_1)}{x_2 - x_1}(t(x_1 - x_2)) \\ &= tf(x_1) + (1-t)f(x_2). \end{aligned}$$

The final statement is the following: For a convex function f on an interval I, over any subinterval $[a, b] \in I$ we have

$$f(at + b(1 - t)) \leq tf(a) + (1 - t)f(b).$$

And this is the statement of Jensen's inequality for two variables. The reverse inequality holds if f is concave, that is if $f''(x) < 0$. This extends to n variables:

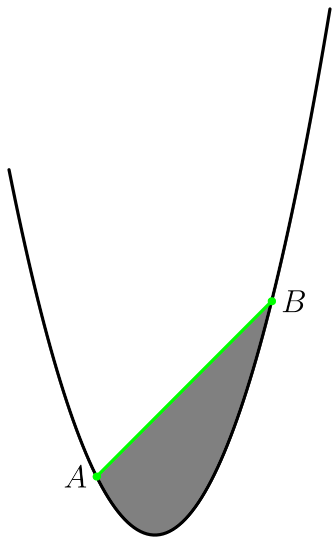

Figure 2: Convex hull of points on $f(x)$ (from A to B)

Theorem 2.1: Jensen's Inequality

Let $f : \mathbb{R} \to \mathbb{R}$ be a convex function over an interval $I \in \mathbb{R}$. Let $x_1, x_2, \ldots x_n \in I$ and positive weights $\lambda_1, \lambda_2, \ldots, \lambda_n > 0$ such that $\sum_{i=1}^n \lambda_i = 1$. Then,

$$f\left(\sum_{i=1}^{n} \lambda_i x_i\right) \leq \sum_{i=1}^{n} \lambda_i f(x_i).$$

The inequality flips signs if f is concave over I.

This could be proved with induction with our previous work as the base case, but I believe there's a more illuminating way to look at it. Notice that if we let $a = x_1$ and $b = x_n$ such that all x_i are contained in the interval $[a, b]$, then every coordinate x_i can be expressed as a linear combination of a and b, so in essence we are still paramterizing the secant line just with more parameters.

First, I'll introduce the idea of a **convex hull**. For a set of points X, the convex hull is the smallest convex set of points containing X. We define convexity for a set of points similarly to functions. A **convex combination** of points x, y is a point $tx + (1-t)y$ with $0 \leq t \leq 1$, or any point on the line segment connecting x and y. A set of points X is defined to be convex if, for any two points x,$y \in X$, every convex combination of x and y is also in X. This is actually the same as our notion for functions, we just think of above the graph as points in the set and below the graph as points not in the set.

The convex hull of points $(x_i, f(x_i))$ is actually precisely the set of points below the secant line from x_1 to x_n and above $f(x)$. So, looking at the statement of Jensen's inequality again:

$$f\left(\sum_{i=1}^{n} \lambda_i x_i\right) \leq \sum_{i=1}^{n} \lambda_i f(x_i).$$

The left hand side f evaluated at some point between x_1 and x_n, which lies on f. Then, the right hand side is a convex combination of points $f(x_i)$ on the graph of f, which must lie within the convex hull, meaning its value is greater than or equal to f evaluated at that point. Instead of using induction, we have delved a bit deeper to explain why the general form of Jensen's inequality holds.

2.1 Problems

Problem 2.1

Let $a, b, c > 0$. Prove that

$$a^a b^b c^c \geq \left(\frac{a+b+c}{3}\right)^{a+b+c}.$$

Problem 2.2

Triangle $\triangle ABC$ has angles A, B, and C. Prove that

$$\sin A + \sin B + \sin C \leq \frac{3\sqrt{3}}{2}.$$

Problem 2.3

Let $a, b, c > 0$ such that $a + b + c = 1$. Prove

$$\left(a + \frac{1}{a}\right)^2 + \left(b + \frac{1}{b}\right)^2 + \left(c + \frac{1}{c}\right)^2 \geq \frac{100}{3}.$$

Problem 2.4: China 1989

Prove that for positive real numbers $x_1, x_2, \ldots x_n$ such that $\sum_{i=1}^n x_i = 1$, the following holds:

$$\sum_{i=1}^{n} \frac{x_i}{\sqrt{1 - x_i}} \geq \frac{\sum_{i=1}^n \sqrt{x_i}}{\sqrt{n-1}}.$$

Problem 2.5: ELMO 2003

Let $x, y, z > 1$ be real numbers such that

$$\frac{1}{x^2 - 1} + \frac{1}{y^2 - 1} + \frac{1}{z^2 - 1} = 1.$$

Prove that

$$\frac{1}{x+1} + \frac{1}{y+1} + \frac{1}{z+1} \leq 1.$$

Problem 2.6: IMO Shortlist 1998

Let $r_1, r_2, \ldots r_n$ be real numbers greater than 1. Prove that

$$\frac{1}{1 + r_1} + \cdots \frac{1}{1 + r_n} \geq \frac{n}{\sqrt[n]{r_1 r_2 \cdots r_n} + 1}.$$

3 AM-GM and Power Mean Inequality

Perhaps the most famous algebraic inequality in competition math is the Arithmetic Mean Geometric Mean (AM-GM) inequality. The two variable statement is as follows:

$$\frac{a+b}{2} \geq \sqrt{ab}$$

There are many ways to prove this, and I'll provide a few. One is based on the trivial inequality, that is $x^2 \geq 0$ for $x \in \mathbb{R}$. Letting $x = a - b$, we have

$$\begin{aligned}(a-b)^2 &\geq 0\\ a^2 + b^2 &\geq 2ab\\ (a+b)^2 &\geq 4ab\\ a+b &\geq 2\sqrt{ab}.\end{aligned}$$

There is also a common geometric proof, specifically a semicircle with diameter $d = a + b$. Then, the radius of the circle is $\frac{a+b}{2}$. Any other point on the semicircle is closer to the diameter, and it can be shown that this has length $\sqrt{ab}$ with similar triangles.

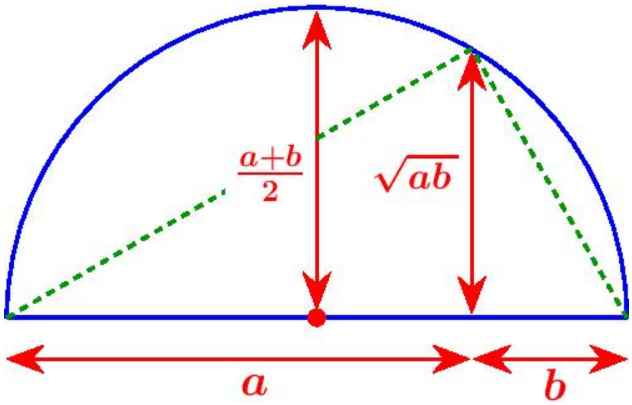

Figure 3: Proof of AM-GM for two variables

Both of these approaches are interesting in their own right, but are not very generalizable. After proving the two variable form, one might just use induction to get the general form of AM-GM, with a lack of insight as to why it's true.

Theorem 3.1: AM-GM

Let $a_1, a_2, \ldots a_n$ be nonnegative reals. Then,

$$\frac{a_1 + a_2 + \cdots a_n}{n} \geq \sqrt[n]{a_1 a_2 \cdots a_n}.$$

The Power Mean Inequality is an extension of AM-GM (you may have heard of QM-AM-GM-HM). The rth power mean (or generalized mean) of a sequence $x_1, x_2, \ldots x_n$ is defined as

$$P(r) = \left(\frac{1}{n}\sum_{i=1}^{n} x_i^r\right)^{\frac{1}{r}}$$

Where $P(0)$ is the geometric mean. This form is similar to something we will get into when exploring Hölder's inequality. A common proof for the Power Mean Inequality (or PMI) is to use induction on AM-GM, but this only covers discrete cases.

Theorem 3.2: Power Mean Inequality

For positive real numbers $\{x_n\}$ and real number r, let

$$P(r) = \left(\frac{1}{n}\sum_{i=1}^{n} x_i^r\right)^{\frac{1}{r}}.$$

Where $P(0)$ is the geometric mean. For real numbers $a, b \in \mathbb{R}$, PMI states that if $a > b$, then $P(a) \geq P(b)$.

3.1 Proofs using Jensen's Inequality

First, let's prove AM-GM with Jensen's inequality. This relies on choosing an appropriate convex/concave function that will get the relationship we want. Starting from Jensen's,

$$f\left(\sum_{i=1}^{n} \lambda_i x_i\right) \leq \sum_{i=1}^{n} \lambda_i f(x_i).$$

The geometric mean requires us to have a product of all x_i terms, which motivates the choice of $f(x) = \ln x$. Now, $f''(x) = -\frac{1}{x^2}$, so f is concave, so we should flip the inequality. Plugging in, we get

$$\ln\left(\sum_{i=1}^{n} \lambda_i x_i\right) \geq \sum_{i=1}^{n} \lambda_i \ln(x_i) = \ln\left(\prod_{i=1}^{n} x_i^{\lambda_i}\right)$$

Now we can exponentiate both sides and preserve the inequality (since $a > b \iff e^a > e^b$), getting

$$\sum_{i=1}^{n} \lambda_i x_i \geq \prod_{i=1}^{n} x_i^{\lambda_i}$$

Setting $\lambda_i = \frac{1}{n}$ for $i = 1, 2, \ldots, n$ gets the familiar form

$$\frac{1}{n}\sum_{i=1}^{n} x_i \geq \sqrt[n]{\prod_{i=1}^{n} x_i}.$$

The approach to prove the Power Mean Inequality is quite similar in nature, focusing on the lens of convex/concave functions with Jensen's, while the introduction of PMI may seem like an unmotivated extension of AM-GM. First, we will prove that $P(t) \geq P(0) \geq P(-t)$ for $t \in \mathbb{R}$, then that $a > b \iff P(a) \geq P(b)$.

From the last step of the AM-GM proof, we can replace all x_i with x_i^t (so we are just changing the sequence we are using Jensen's with $f(x) = \ln x$ on) to get

$$\sum_{i=1}^{n} \lambda_i x_i^t \geq \prod_{i=1}^{n} x_i^{\lambda_i t} \implies \left(\sum_{i=1}^{n} \lambda_i x_i^t\right)^{\frac{1}{t}} \geq \prod_{i=1}^{n} x_i^{\lambda_i} = P(0).$$

So $P(t) \geq P(0)$. Identical steps can be taken to show $P(-t) \leq P(0)$, since in the final step raising each side to the power of $-\frac{1}{t}$ would cause the inequality to flip.

Now for the second part of the proof. Let a, b be reals such that $a > b$ and $ab > 0$ (so either $a, b > 0$ or $a, b < 0$). Then, we choose the function $f(x) = x^{\frac{a}{b}}$ and apply Jensen's to the sequence $\{x^b\}_{i=1}^n$. We have $f''(x) = \left(\frac{a}{b}\right)\left(\frac{a}{b} - 1\right) x^{\frac{a}{b}-2}$ which is greater than 0 since $\frac{a}{b} > 1$, so $f(x)$ is convex. Applying Jensen's, we get

$$\left(\sum_{i=1}^n \lambda_i x_i^b\right)^{\frac{a}{b}} \leq \sum_{i=1}^n \lambda_i x_i^a \implies \left(\sum_{i=1}^n \lambda_i x_i^b\right)^{\frac{1}{b}} \leq \left(\sum_{i=1}^n \lambda_i x_i^a\right)^{\frac{1}{a}}.$$

So $P(b) \leq P(a)$. Any cases where a and b are different signs are resolved by the first part of the proof, since each are bounded above/below by $P(0)$. Thus, we have proved AM-GM and the Power Mean Inequality using Jensen's, which has also let us use arbitrary weights.

Another interesting connection is that we always use a function of the form x^r for proving the Power Mean Inequality, but end up using $\ln x$ when comparing to the geometric mean. This is weirdly similar to differentiation/integration of x^r and how $\ln x$ serves as a 'bridge' between positive and negative powers of x. The geometric mean is obtained as $P(0)$ as a limiting case from the definition, and the proof for this in fact depends on the definition of $\ln x$. So, AM-GM and PMI are simply the consequence of a property of convex/concave functions.

3.2 Problems

Problem 3.1: Young's Inequality

Let $a, b > 0$ such that $\frac{1}{a} + \frac{1}{b} = 1$. Prove that

$$xy \leq \frac{x^a}{a} + \frac{x^b}{b}.$$

Problem 3.2

Let $a, b, c > 0$. Prove that

$$\frac{a^2 + b^2 + c^2}{abc} \geq \frac{1}{a} + \frac{1}{b} + \frac{1}{c}.$$

Problem 3.3

Let $a, b, c > 0$. Prove that

$$\frac{a^3}{bc} + \frac{b^3}{ac} + \frac{c^3}{ab} \geq a + b + c.$$

Problem 3.4

Let a, b, c be positive reals such that $a + b + c = 3$. Prove that

$$a^b b^c c^a \leq 1.$$

Hint: Weighted AM-GM.

Problem 3.5: Nesbitt's Inequality

For $a, b, c > 0$, prove that

$$\frac{a}{b+c} + \frac{b}{a+c} + \frac{c}{a+b} \geq \frac{3}{2}.$$

Problem 3.6: Taiwan TST 2014

Let a, b, c be positive reals. Prove that

$$3(a+b+c) \geq 8\sqrt[3]{abc} + \sqrt[3]{\frac{a^3+b^3+c^3}{3}}.$$

Hint: Power Mean Inequality.

4 Sidenotes

4.1 Muirhead and Karamata

I will present the statements of Muirhead and Karamata's (which is an extension of Muirhead's to functions). Often Karamata and Jensen are presented together as we move into the realm of considering functions in general over simple algebraic inequalities. First, we define majorization in sequences.

Definition 4.1: Majorization

A sequence $\{x_n\}$ majorizes another sequence $\{y_n\}$ if, for all $1 \leq i < n$, we have

$$\sum_{k=1}^{i} x_k \geq \sum_{k=1}^{i} y_k$$

and

$$\sum_{k=1}^{n} x_k = \sum_{k=1}^{n} y_k.$$

Theorem 4.1: Muirhead's Inequality

Let $\{a_n\}$ be a sequence of nonnegative real numbers, and let $\{x_n\}$ majorize $\{y_n\}$. Then,

$$\sum_{sym} a_1^{x_1} a_2^{x_2} \cdots a_n^{x_n} \geq \sum_{sym} a_1^{y_1} a_2^{y_2} \cdots a_n^{y_n}.$$

Muirhead's generalizes repeated applications of AM-GM to inequalities. For example, the inequality

$$a^2 + b^2 + c^2 \geq ab + bc + ac$$

is true since $(2,0,0) \succ (1,1,0)$. Muirheads is a symmetric sum (sum over all permutations), so it would get that

$$2a^2 + 2b^2 + 2c^2 \geq 2ab + 2bc + 2ac \implies a^2 + b^2 + c^2 \geq ab + bc + ac.$$

And, as we have seen, this can be shown with three applications of AM-GM on pairs of terms on the left side.

Theorem 4.2: Karamata's Inequality

Let $f(x)$ be a convex function over an interval $I \in \mathbb{R}$. Suppose sequences $\{x_n\}$,$\{y_n\} \in I$ such that $\{x_n\}$ majorizes $\{y_n\}$. Then,

$$f(x_1) + f(x_2) + \cdots f(x_n) \geq f(y_1) + f(y_2) + \cdots f(y_n).$$

Karamata's inequality can be looked at as a more general Jensen's inequality, just as Muirheads is in a way a more general AM-GM. Consider sequences $\{a_n\}$ and $\{b_n\}$ such that $\{b_n\} \succ \{a_n\}$. Then, $\{a_n\}$ is, in a sense, a more 'spread out' (or smoothed out) version of $\{b_n\}$, so we can express terms of $\{a_n\}$ as convex combinations of terms of $\{b_n\}$. That is, for all $1 \leq i \leq n$,

$$a_i = \sum_{j=1}^{n} \lambda_{ij} b_j$$

Where $\sum_j \lambda_{ij} = \sum_i \lambda_{ij} = 1$ (The λ_{ij} are values in a doubly stochastic matrix). Then, consider a convex function f. By Jensen's,

$$f(a_i) = f\left(\sum_{j=1}^{n} \lambda_{ij} b_j\right) \leq \sum_{j=1}^{n} \lambda_{ij} f(b_j)$$

Now we sum both sides over all i:

$$\sum_{i=1}^{n} f(a_i) \leq \sum_{i=1}^{n} \sum_{j=1}^{n} \lambda_{ij} f(b_j) = \sum_{j=1}^{n} f(b_j) \sum_{i=1}^{n} \lambda_{ij} = \sum_{j=1}^{n} f(b_j).$$

Therefore, we have shown that $\{b_n\} \succ \{a_n\}$ implies

$$\sum_{i=1}^{n} f(a_i) \leq \sum_{j=1}^{n} f(b_j).$$

We can use Karamata's to get a quick proof of Muirhead's inequality, if desired. There are also some interesting linear algebraic ideas behind proving Muirhead's when the exponents can be any real number, but I won't get into that in this lesson.

4.2 Homogenization

Homogenization of an inequality is a common strategy when given conditions on your variables (such as $a+b+c=1$) and the degree of terms in inequality don't match up. For example, consider if I asked you to prove (for $a, b, c > 0$):

$$a + b + c \geq 3.$$

Obviously this is not always true, and we can easily pick values of a, b, c that don't satisfy this inequality and values that do. However, if I imposed the condition $abc = 1$, we would see that the inequality

$$a + b + c \geq 3\sqrt[3]{abc} = 3$$

is true by AM-GM. Note that the degree of both sides of the dehomogenized inequality is 1, since all terms $(a, b, c, \sqrt[3]{abc})$ have degree 1. Generally, in a homogeneous inequality we can arbitrarily scale $(a, b, c) \to (ka, kb, kc)$ and it does not change the inequality. This is why homogenization is used to eliminate conditions imposed on inequalities. As I did above, inserting $\sqrt[3]{abc}$ on the right hand side makes the condition $abc = 1$ irrelevant, because we could have $abc = k$ by multiplying a, b, c by some factor, and after the insertion it does not change the statement of the inequality. Note that all of our fundamental inequalities (AM-GM, Power Mean, Cauchy) are homogeneous, and are true for all positive inputs.

Problem 4.1

Let $a, b, c > 0$ such that $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1$. Prove that

$$(a+1)(b+1)(c+1) \geq 64.$$

This problem can be used as a homogenization exercise. There are many different solutions that inherently homogenize the inequality, for example we can homogenize terms using PMI and comparing to the Power Mean Inequality to get a degree n comparison to degree 0. Otherwise this is just an exercise in algebraic manipulation, to some extent.

5 Cauchy and Hölder

Consider two vectors in $\mathbb{R}^n$, $\vec{x} = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix}$ and $\vec{y} = \begin{bmatrix} y_1 & y_2 & \cdots & y_n \end{bmatrix}$. The magnitude of the dot product of these vectors is $|\vec{x}||\vec{y}|\cos\theta$, where $\cos\theta$ is the angle between the two vectors. Since $\cos\theta \leq 1$, this fact gives us the famous Cauchy-Schwarz inequality:

Theorem 5.1: Cauchy-Schwarz Inequality

Let $\vec{x} = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix}$ and $\vec{y} = \begin{bmatrix} y_1 & y_2 & \cdots & y_n \end{bmatrix}$ be real valued vectors in $\mathbb{R}^n$. Then,

$$\vec{x} \cdot \vec{y} \leq |\vec{x}||\vec{y}|.$$

Which can be written algebraically as

$$x_1y_1 + x_2y_2 + \cdots x_ny_n \leq \sqrt{(x_1^2 + x_2^2 + \cdots x_n^2)(y_1^2 + y_2^2 + \cdots y_n^2)}.$$

Cauchy is one of my favorite algebraic inequalities to use. However, we will be focusing on a different aspect of it (besides solving inequalities) that extends to function spaces with different norms. Essentially, the Euclidean norm used in Cauchy-Schwarz provides a way to control the inner product (dot product) of the vectors.

Here we'll shift the focus of inequalities in this article. Before, we were exploring algebraic inequalities (mainly through Jensen's) with the purpose of applying them to proving inequalities. From now on, I won't add many exercises, because instead of problem solving the focus is now understanding how we can control function behavior.

The Euclidean norm is most familiar as what we use to measure the distance between points in $\mathbb{R}^n$. First, I'll provide the requirements for a function $f : V \to [0, \infty)$ on a vector space V to be a *norm*.

Definition 5.1: Norm

A function $||\cdot|| : V \to [0, \infty)$ is a norm if it satisfies all of the following properties for $x, y \in V$ and $\alpha \in \mathbb{R}$:

1. Positive Definiteness: $||x|| \leq 0$ and $||x|| = 0 \iff x = 0$
2. Absolute Homogeneity: $||\alpha x|| = |\alpha| ||x||$
3. Triangle Inequality: $||x + y|| \leq ||x|| + ||y||$

It's fairly obvious that the Euclidean norm satisfies these properties. I will now introduce the p-norm, which is very similar to the generalized mean.

Definition 5.2: p-norm

For real $p \geq 1$, the p-norm of a real valued vector $\vec{x} = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix}$ is defined as

$$||x||_p = \left(\sum_{i=1}^{n} |x_i|^p \right)^{\frac{1}{p}}.$$

For some intuition, try to imagine (or draw) what the equation $||x||_p = 1$ looks like in $\mathbb{R}^2$ for certain values of p. Then, verify that this is a norm. The harder part is likely the triangle inequality. For p-norms, this is also called Minkowski's Inequality:

Theorem 5.2: Minkowski's Inequality

Triangle Inequality analog for p-norms. That is,

$$||x + y||_p \leq ||x||_p + ||y||_p$$

Which can also be stated as

$$\left(\sum_{i=1}^{n} |x_i + y_i|^p \right)^{\frac{1}{p}} \leq \left(\sum_{i=1}^{n} |x_i|^p \right)^{\frac{1}{p}} + \left(\sum_{i=1}^{n} |y_i|^p \right)^{\frac{1}{p}}.$$

Consider $f(x) = |x|^p$. For $p \geq 1$, this function is convex, so we can apply Jensen's inequality for two variables:

$$|x + y|^p = \left| (1 - t)\frac{x^p}{1 - t} + t\frac{y^p}{t} \right| \leq \frac{|x|^p}{(1 - t)^{p-1}} + \frac{|y|^p}{t^{p-1}}.$$

This is true for any x, y, so we can sum to get closer to the desired form:

$$\sum_{i=1}^{n} |x_i + y_I|^p \leq \frac{1}{(1 - t)^{p-1}} \sum_{i=1}^{n} |x_i|^p + \frac{1}{t^{p-1}} \sum_{i=1}^{n} |y_i|^p.$$

Note that this is now equivalent to

$$||x + y||_p^p \leq \frac{||x||_p^p}{(1 - t)^{p-1}} + \frac{||y||_p^p}{t^{p-1}}.$$

Finally, we choose $t = \frac{||y||_p}{||x||_p + ||y||_p}$ to get the desired inequality. If $p < 1$, then the p-norm is no longer a norm since $f(x) = |x|^p$ is no longer convex. $\square$

Now that we have verified that the p-norm is a norm, we can state Hölder's Inequality, which is Cauchy-Schwarz generalized from the Euclidean norm to p-norms.

Theorem 5.3: Hölder's Inequality

Let $p, q \in (1, \infty)$ such that $\frac{1}{p} + \frac{1}{q} = 1$, and real (or complex) valued vectors $\vec{x}, \vec{y} \in \mathbb{R}^n$. Then,

$$\vec{x} \cdot \vec{y} \leq ||x||_p ||y||_q.$$

Algebraically, this is

$$\sum_{i=1}^{n} x_i y_i \leq \left(\sum_{i=1}^{n} |x_i|^p \right)^{\frac{1}{p}} \cdot \left(\sum_{i=1}^{n} |y_i|^q \right)^{\frac{1}{q}}$$

Note that $p = q = 2$ gets the Cauchy-Schwarz Inequality.

If you want a hint for proving this, consider Problem 3.1 (Young's Inequality). In a way, Cauchy is the tightest upper bound on the inner product obtained from Hölder's, and changing p and q give a little more leeway, but we will get more into what *control* means here later.

Now for the proof. We have $\vec{x} = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix}$ and $\vec{y} = \begin{bmatrix} y_1 & y_2 & \cdots & y_n \end{bmatrix}$. Consider $\vec{a} = \frac{\vec{x}}{||x||_p}$ and $\vec{b} = \frac{\vec{y}}{||y||_p}$. Young's Inequality tells us, for each $1 \leq i \leq n$,

$$a_i b_i \leq \frac{a_i^p}{p} + \frac{b_i^q}{q}.$$

Summing and multiplying by $||x||_p ||y||_q$, this gets

$$\sum_{i=1}^{n} |x_i y_i| \leq ||x||_p ||y||_q \sum_{i=1}^{n} \left(\frac{a_i^p}{p} + \frac{b_i^q}{q} \right) = ||x||_p ||y||_q \implies \vec{x} \cdot \vec{y} \leq ||x||_p ||y||_q.$$

With the last equality since we have defined $\vec{a}$ and $\vec{b}$ to have p-norm 1 and $\frac{1}{p} + \frac{1}{q} = 1$. Equality in Young's Inequality holds when $a^p = b^q$, which in this proof translates to x_i^p being a multiple of y_i^q for all i. $\square$

6 L^p spaces

The meaning and usefulness of p-norms and Hölder's inequality comes once we extend the scope of functions we are considering. Formally, this requires measure theory, but we can stick with real valued functions and still see the usefulness of these ideas. We extend sequences to functions, and discrete sums become integrals over measures. Our p-norm becomes

$$||f||_p = \left(\int_{\Omega} |f(x)|^p d\mu(x) \right)^{\frac{1}{p}}.$$

For a measure $\mu(x)$ and domain Ω. The measure pretty much generalizes assignment of size, but is now applicable to broader groups of sets (not just nice ones like $\mathbb{R}$ and Riemann integration). However, we will just deal with Riemann integration in these examples for simplicity. So we can define L^p spaces:

Definition 6.1: L^p space

A L^p space contains all functions that are measurable (for a fixed measure (X, M, μ)) and have a finite $p-$norm. That is.

$$L^p = \{f : X \to \mathbb{C} : f \text{ is measurable and } ||f||_p < \infty\}.$$

When only considering $\mathbb{R}$, to have $f \in L^p$ all we need is

$$||f||_p = \left(\int_\Omega |f(x)|^p dx\right)^{\frac{1}{p}} < \infty$$

And since $p \neq 0$, we just require the integral of $|f|^p$ over some domain $\Omega \in \mathbb{R}$ to be finite.

L^p spaces are a tool used for identifying the regularity or behavior of functions. Since the p-norm is finite, the growth/decay of the function is controlled. Generally, a larger p indicates a stronger level of control over the behavior of the function (consider L^∞) since it is harder to remain finite. In general, for functions on *finite domains*, we have that $1 < p < q \implies L^q \subset L^p$. This is not generally true for infinite domains.

First, a few examples. L^p spaces only really give more usefulness for functions that exhibit some sort of asymptotic decay, and p is a way to measure the rate of that decay in a meaningful way. Consider $f = \frac{1}{x^a}$ fo $a > 0$ on $[1, \infty)$. Then, we see

$$||f||_p^p = \int_1^\infty \frac{1}{x^{ap}} dx$$

This is only finite if $ap > 1$, or $p > \frac{1}{a}$. So, we'd say $f \in L^p$ for $p > \frac{1}{a}$.

Now, our domain matters. Consider the same function f on $(0, 1]$. Then,

$$||f||_p^p = \int_0^1 \frac{1}{x^{ap}} dx$$

However, this only converges when $ap < 1$. So now $f \in L^p$ for any $p < \frac{1}{a}$. L^p spaces value functions 'blowing up' and 'decaying' differently. In general, as integration is accumulation, we are measuring the 'accumulation to the pth power'. If a function is accumulating too much, on average, then its p-norm will diverge. This could mean it is too slow to decay, or that it blows up too fast, so we need smaller p to capture the function. Generally, then, higher p means a more controlled decay/growth. Notice that these conditions for $\frac{1}{x^a}$ are contradictory, so we actually can't have $f \in L^p(0, \infty)$ for any p, which can be visualized as changing a results in a tradeoff of decay and blowing up (we can see this graphically, as these terms are far from formal).

Making this extension, we also get a continuous version of Hölder's inequality.

Theorem 6.1: Hölder's Inequality (L^p)

Let $p, q \in (1, \infty)$ such that $\frac{1}{p} + \frac{1}{q} = 1$. Let $f \in L^p$ and $g \in L^q$. Then,

$$||fg||_1 \leq ||f||_p ||g||_q.$$

This is an example of where the *control* is. Let's say we had some weird function $f \in L^p$, we could control it using a function $g \in L^q$, such that the product $fg \in L^1$. If we have two functions

f and g that behave in L^p and L^q space, then their product is bounded in L^1 by Hölder's. As an example, consider $f(x) = x^{-\frac{1}{2}}$ and $g(x) = e^{-x}$ on $[1, \infty)$.

$$||f||_p^p = \int_1^\infty \frac{1}{x^{\frac{p}{2}}} dx$$

Let's choose $p = 4$ since we see that will make $|f|^p$ finite. The conjugate exponent q is then $\frac{4}{3}$, and we can check

$$||g||_q^q = \int_1^\infty e^{-\frac{4x}{3}} dx$$

Which is also finite. That means we now know that

$$\int_1^\infty x^{-\frac{1}{2}} e^{-x} dx \le \left(\int_1^\infty \frac{1}{x^2} dx \right)^{\frac{1}{4}} \left(\int_1^\infty e^{-\frac{4x}{3}} dx \right)^{\frac{3}{4}} = \frac{1}{e} \left(\frac{3}{4} \right)^{\frac{3}{4}}.$$

Now isn't that pretty cool. The integral on the left actually doesn't have a closed form (it's related to the error function), so it's pretty interesting that we can bound it with Hölder's. Experiment with different p values and see if you can prove that Cauchy is the best bound!

7 Conclusion

Thanks to everyone in Senior Math Team for listening to me yap for three weeks about this, and for everyone who read through the paper! Honestly, inequalities can get pretty crazy when you start to see how far they are generalized, and I think that's pretty cool, but it's something I'm not able to teach as I am now. If anyone notices any issues with the document, please let me know!

8 Problem Solutions

Solutions to Selected Problems below. I didn't feel like writing sols to all of them, but the interesting ones are here.

Problem 1

Let $a, b, c > 0$. Prove that

$$a^a b^b c^c \ge \left(\frac{a+b+c}{3} \right)^{a+b+c}.$$

Solution 1. First, take the natural log of both sides:

$$a \ln a + b \ln b + c \ln c \ge (a+b+c) \ln \left(\frac{a+b+c}{3} \right).$$

This motivates choosing $f(x) = x \ln x$ and applying Jensen's with weights $\lambda_i = \frac{1}{3}$. We have $f''(x) = \frac{1}{x}$, so f is convex for all $x > 0$. By Jensen's, we have

$$\begin{gathered}
\frac{1}{3}(f(a) + f(b) + f(c)) \ge f\left(\frac{a+b+c}{3} \right) \\
\frac{1}{3}(a \ln a + b \ln b + c \ln c) \ge \frac{a+b+c}{3} \ln \left(\frac{a+b+c}{3} \right) \\
\implies a \ln a + b \ln b + c \ln c \ge (a+b+c) \ln \left(\frac{a+b+c}{3} \right).
\end{gathered}$$

Which can be rearranged to the original problem statement. □

Problem 2

Triangle $\triangle ABC$ has angles $A, B,$ and C. Prove that

$$\sin A + \sin B + \sin C \leq \frac{3\sqrt{3}}{2}.$$

Solution 2. The important information we have is that, since $\triangle ABC$ is a triangle, $A+B+C = 180^\circ$. Now, we consider Jensen's with equal weights on $f(x) = \sin x$. Since $0 < A, B, C < \pi$, and $f''(x) = -\sin x$ is concave on $I = (0, \pi)$, we can apply Jensen's to get

$$\frac{1}{3}(\sin A + \sin B + \sin C) \leq \sin\left(\frac{A+B+C}{3}\right) = \sin 60^\circ.$$

Which rearranges to the desired

$$\sin A + \sin B + \sin C \leq \frac{3\sqrt{3}}{2}.$$

□

Problem 3

Let $a, b, c > 0$ such that $a + b + c = 1$. Prove

$$\left(a + \frac{1}{a}\right)^2 + \left(b + \frac{1}{b}\right)^2 + \left(c + \frac{1}{c}\right)^2 \geq \frac{100}{3}.$$

Solution 3. There are multiple ways to go about proving this, with varying algebraic manipulation. First, we can expand on the left side fairly nicely and rearrange to get

$$a^2 + \frac{1}{a^2} + b^2 + \frac{1}{b^2} + c^2 + \frac{1}{c^2} \geq \frac{82}{3}$$

To prove this, we consider Jensen's on $f(x) = x^2 + \frac{1}{x^2}$. We can see that $f''(x) = 2 + \frac{6}{x^4}$, which is always positive so f is convex. Therefore, we have that

$$\begin{aligned} \frac{1}{3}(f(a) + f(b) + f(c)) &\geq f\left(\frac{a+b+c}{3}\right) = f\left(\frac{1}{3}\right) \\ \implies a^2 + \frac{1}{a^2} + b^2 + \frac{1}{b^2} + c^2 + \frac{1}{c^2} &\geq 3f\left(\frac{1}{3}\right) = 3 \cdot \frac{82}{9} = \frac{82}{3}. \end{aligned}$$

□

Problem 4: China 1989

Prove that for positive real numbers $x_1, x_2, \ldots x_n$ such that $\sum_{i=1}^n x_i = 1$, the following holds:

$$\sum_{i=1}^n \frac{x_i}{\sqrt{1-x_i}} \geq \frac{\sum_{i=1}^n \sqrt{x_i}}{\sqrt{n-1}}.$$

Solution 4. Motivated by the sum on the left side, we consider $f(x) = \frac{x}{\sqrt{1-x}}$ with equal weights. Notice that since all x_i are positive and sum to 1, $f(x)$ is viable since $f''(x) = (1-x)^{-\frac{3}{2}} + \frac{3}{4}x(1-x)^{-\frac{5}{2}}$ is positive for $x \in (0,1)$, so f is convex over this interval. Jensen's gets us

$$\frac{1}{n}\sum_{i=1}^{n}\frac{x_i}{\sqrt{1-x_i}} \geq f\left(\frac{1}{n}\right) = \frac{1}{\sqrt{n^2-n}}.$$

Rearranging, we get

$$\sum_{i=1}^{n}\frac{x_i}{\sqrt{1-x_i}} \geq \frac{\sqrt{n}}{\sqrt{n-1}}.$$

This is very close to what we want. We can show that $\sqrt{n} \geq \sum_{i=1}^{n}\sqrt{x_i}$ directly from Cauchy-Schwarz on $\vec{a} = \begin{bmatrix}\sqrt{x_1} & \sqrt{x_2} & \dots \sqrt{x_n}\end{bmatrix}$ and the ones vector of length n. So, we have

$$\sum_{i=1}^{n}\frac{x_i}{\sqrt{1-x_i}} \geq \frac{\sqrt{n}}{\sqrt{n-1}} \geq \frac{\sum_{i=1}^{n}\sqrt{x_i}}{\sqrt{n-1}}.$$

□

Problem 5: ELMO 2003

Let $x, y, z > 1$ be real numbers such that

$$\frac{1}{x^2-1} + \frac{1}{y^2-1} + \frac{1}{z^2-1} = 1.$$

Prove that

$$\frac{1}{x+1} + \frac{1}{y+1} + \frac{1}{z+1} \leq 1.$$

Solution 5. I didn't give this problem in class because I thought the Jensen's solution kind of sucked, but there are other ways too (namely Cauchy). Basically, we let $a = \frac{1}{x^2-1}$ and similarly for $b \to y$ and $c \to z$. Then, $x = \sqrt{\frac{1}{a}+1}$, and we'd like to show

$$\frac{1}{\sqrt{\frac{1}{a}+1}+1} + \frac{1}{\sqrt{\frac{1}{b}+1}+1} + \frac{1}{\sqrt{\frac{1}{c}+1}+1} \leq 1.$$

This looks quite ugly, but considering the (very motivated) use of Jensen's on $f(x) = \frac{1}{1+\sqrt{\frac{1}{x}+1}}$, which one can verify to be concave on their own time, we immediately get the desired inequality. Try to find the Cauchy solution!

Problem 6: IMO Shortlist 1998

Let $r_1, r_2, \dots r_n$ be real numbers greater than 1. Prove that

$$\frac{1}{1+r_1} + \cdots \frac{1}{1+r_n} \geq \frac{n}{\sqrt[n]{r_1 r_2 \cdots r_n}+1}.$$

Solution 6. The left side motivates a function of the form $f(x) = \frac{1}{1+x}$, but that doesn't quite get the geometric mean of terms on the right side (instead an arithmetic mean). Drawing some inspiration from the AM-GM proof, we want something exponential based instead of x. Consider $f(x) = \frac{1}{1+e^x}$, and notice $f''(x) = \frac{e^x(e^x-1)}{(1+e^x)^3}$ is positive when $e^x > 1$. Note that now we are setting $e^{x_i} = r_i > 1$, and applying Jensen's with equal weights gets

$$\frac{1}{1+e^{x_i}} + \cdots + \frac{1}{1+e^{x_n}} \geq \frac{n}{e^{\frac{x_1+\cdots x_n}{n}}+1}.$$

Since $e^{\frac{x_1+\cdots x_n}{n}} = (e^{x_1}e^{x_2}\cdots e^{x_n})^{\frac{1}{n}} = \sqrt[n]{r_1 r_2 \cdots r_n}$, once we substitute back $e^{x_i} \to r_i$ we are done. □

Problem 7: Young's Inequality

Let $a, b > 0$ such that $\frac{1}{a} + \frac{1}{b} = 1$. Prove that

$$xy \leq \frac{x^a}{a} + \frac{x^b}{b}.$$

Solution 7. This is a direct result of weighted AM-GM. We apply weighted AM-GM on x^a and x^b with weights $\frac{1}{a}$ and $\frac{1}{b}$, getting

$$(x^a)^{\frac{1}{a}}(y^b)^{\frac{1}{b}} \leq \frac{x^a}{a} + \frac{x^b}{b}.$$

And since the left side equals xy, we are done. □

Problem 8

Let a, b, c be positive reals such that $a + b + c = 3$. Prove that

$$a^b b^c c^a \leq 1.$$

Solution 8. Motivated with Weighted AM-GM, we want to weight a by b, b by c, and c by a. Using this, we know that

$$\frac{ab + bc + ac}{a + b + c} \geq a^b b^c c^a.$$

Now we can notice that $a^2 + b^2 + c^2 \geq ab + ac + bc$ by repeated AM-GM or Muirhead's (this fact is used enough to be well known), rewriting as $(a + b + c)^2 \geq 3(ab + ac + bc)$, we can rearrange and use $a + b + c = 3$ to get

$$1 \geq \frac{ab + bc + ac}{a + b + c}.$$

Substituting this back into the inequality above gets us the desired form. □

Problem 9: Taiwan TST 2014

Let a, b, c be positive reals. Prove that

$$3(a + b + c) \geq 8\sqrt[3]{abc} + \sqrt[3]{\frac{a^3 + b^3 + c^3}{3}}.$$

Solution 9. We notice that each of the terms look similar to a mean (arithmetic, geometric, $P(3)$). First, we divide both sides by 9:

$$\frac{a + b + c}{3} \geq \frac{8}{9}\sqrt[3]{abc} + \frac{1}{9}\sqrt[3]{\frac{a^3 + b^3 + c^3}{3}}.$$

We can use Power Mean to 'get rid' of the cube roots in the terms. In particular, we use the weighted power mean with $P(\frac{1}{3}) \leq P(1)$, getting

$$\left(\frac{8}{9}\sqrt[3]{abc} + \frac{1}{9}\sqrt[3]{\frac{a^3+b^3+c^3}{3}}\right)^3 \leq \frac{8}{9}(abc) + \frac{1}{9}\left(\frac{a^3+b^3+c^3}{3}\right) = \frac{a^3+b^3+c^3+24abc}{27}$$

Now all we need to do is show that

$$\left(\frac{a+b+c}{3}\right)^3 \geq \frac{a^3+b^3+c^3+24abc}{27},$$

which can be verified by expanding the left hand side and using AM-GM/Muirhead's. $\square$

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